SUPPORTING ADULT GUIDANCE

One number, many representations

Place Value · Lesson 8. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

In the main activity, the Place-Value Museum, one number appears in six linked display cases: digits, words, a place-value chart, a number line, money (£3.45 = 345p) and length (3.45 m = 345 cm = 3,450 mm). Changing the number in any case updates all the others, and the case just changed is outlined. Eight curator challenges then show one representation and ask for another (for example £2.07 in pence). A stretch explains why £3.45 is 345p. Learners then switch quickly between units, choose which representation suits a job, and build exhibits that meet conditions. This lesson also prepares for the geometry units, where lengths in metres and centimetres are read and compared.

Driving Question: *Which way of showing this number helps most?*

The mathematics, in plain terms

Pence are hundredths of a pound and centimetres are hundredths of a metre, so converting is multiplying or dividing by 100 (Lesson 2). Answers are accepted with or without units, and in either unit of a pair (207p or £2.07). Bare numbers are read in the unit the question asks for. The museum works to hundredths so that money and centimetres stay whole; millimetres (thousandths of a metre) appear only in the length display.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — £3.45 in pence

How many pence is £3.45?

£3.45

Ones (£1)		Tenths (10p)	Hundredths (1p)
3	•	4	5

Pence are hundredths of a pound: the 5 is 5p, the 4 is 40p, the 3 is 300p.

1. 3 pounds is 300p
2. 4 tenths of a pound is 40p
3. 5 hundredths of a pound is 5p
4. Altogether 345p

£3.45 = 345p. The digits keep their order; the unit changes.

Example 2 — 1.5 m in centimetres

How many centimetres is 1.5 m?

1.5 m

Ones (1 m)		Tenths (10 cm)	Hundredths (1 cm)
1	•	5	0

1.5 m = 1.50 m. The 0 holds the hundredths (centimetres) column.

1. 1 metre is 100 cm
2. 5 tenths of a metre is 50 cm
3. Altogether 150 cm

1.5 m = 150 cm, not 15 cm.

See all six at once, then convert, then choose

Learners see the representations change together before converting on their own, so conversion is a matter of reading another case rather than following a rule. The stretch explains the shared digits. Choosing the best representation, and building exhibits under conditions, close the unit.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The Place-Value Museum and eight curator challenges. Stretch: why is £3.45 the same as 345p?	~16 min
Movement	Quick switch: pounds, pence, metres and centimetres.	~7 min
Reflection	Which helps most? Choose a representation for a job.	~8 min
Creativity	Build an exhibit that fits a label.	~10 min
Rest	Same number, different clothes.	~2 min
Nutrition	Intellectual: choice.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Writing £2.07 as 27p or 270p.
Ask: “Which column is the 0 in? What does it hold?”

Writing 1.5 m as 15 cm.
Ask: “How many centimetres in one metre? So in half a metre?”

Writing £3.5 for three pounds fifty.
Ask: “How many digits do we use for pence? What does the 5 stand for?”

Comparing 0.8 m and 75 cm by the digits (75 > 8).
Ask: “Can you write both in centimetres first?”

Thinking words show size well (four point one five sounds bigger than four point two).
Ask: “Which representation would show which is larger?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the whole Place Value unit (Lessons 1 to 7). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says 85p is £8.50. The aim is a question that lets them check with the chart.

- “How many pence make £1?”
- “Is 85p more or less than £1?”

- “In the money case, which column do pence go in?”
- “Can you try it in the museum and read the money case?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Should learners include units in their answers?
Encourage it: “207p” is clearer than “207”. The lesson accepts answers with or without units, and in either unit (207p or £2.07).

Why measures in a place-value unit?
The next units are geometry; reading and converting lengths is the place-value idea in use, and it will be needed straight away.

Is the stretch required?
No. The secure core is converting between pounds and pence and between metres and centimetres, and explaining why the digits keep their order.

What comes next?
This is the last lesson of the unit. The next unit on the route is Angles Tell Us Why, with Perimeter and Area.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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